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YouTube Videos Analysis

1 Introduction

Uncovering the Secrets of YouTube Trending Videos

The mission is to explore the YouTube trending videos dataset and uncover the key factors behind viral content. Through exploratory data analysis (EDA) and data visualization, we will analyze engagement patterns, trends, and content characteristics. However, this is not just about answering a set of predefined questions—we need to craft your own data-driven story and develop meaningful insights based on it.

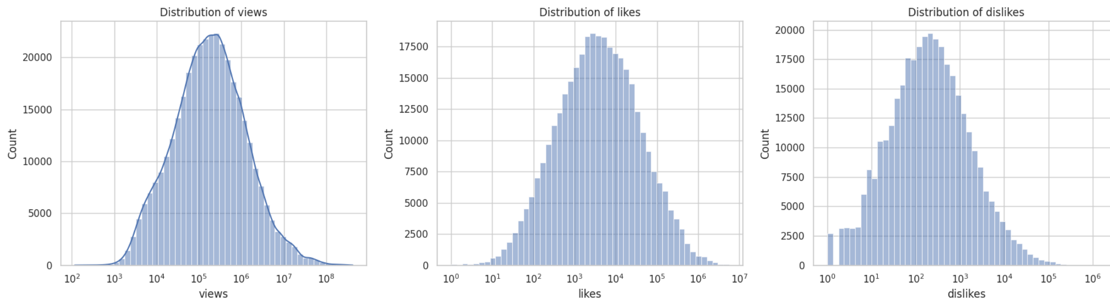
All trending videos aren't created equal—some explode, others fizzle. What patterns can we find that explain why certain trending videos perform far better than others?

Since all videos are trending, we define virality using a measurable threshold: Top 10% in engagement ratio

and then we answer these questions:

1. How are engagement metrics (views, likes and dislikes) distributed overall and across different video categories?
2. Which YouTube channels and video categories trend the most in each country and globally?
3. Are there seasonal or day-of-week patterns in trending videos? How does the upload day and time impact video engagement?
4. Do controversial videos, defined by a high dislike ratio, receive more engagement than universally liked ones?
5. How do video tags influence engagement, and which tags are most commonly used in trending videos?
6. How does the length of a video title impact engagement levels?
7. Is there a relationship between video title sentiment, whether positive, neutral, or negative, and engagement levels? (extra point)
8. Do clickbait-style titles, such as those containing words like "shocking" or "must watch," result in higher engagement? (extra point)
9. How does the title sentiment category (positive, neutral, negative) differ between viral and non-viral trending videos?
10. What is the effect of number of tags (num_tags) on virality? Is more always better?
11. Does publish hour matter? Are viral videos posted at specific times of day?
12. Do clickbait-style titles help more in some categories than others?

13. Which channels appear most frequently among viral trending videos? Do they post more often or just better content?
14. Are there cross-country differences in what goes viral?



2 EDA

2.1 Dataset Overview

- The dataset consists of **363372 rows** and **18 columns**.
- In fact, we have 10 datasets from 10 countries, and 10 JSON file for each.
- The countries are: USA, Great Britain, Germany, Canada, France, Russia, Mexico, South Korea, Japan, India

Column	Type	Description
`video_id`	object	Unique video identifier
`trending_date`	datetime64[ns]	Date the video started trending
`title`	object	Title of the video
`channel_title`	object	Name of the channel
`category_id`	int64	ID representing the category of the video
`publish_time`	datetime64[ns]	Time the video was published
`tags`	object	Comma-separated tags associated with the video
`views`	int64	Number of views the video received
`likes`	int64	Number of likes the video received
`dislikes`	int64	Number of dislikes the video received
`comment_count`	int64	Number of comments on the video

Column	Type	Description
`thumbnail_link`	object	URL link to the video thumbnail
`comments_disabled`	bool	Whether comments are disabled on the video
`ratings_disabled`	bool	Whether ratings are disabled on the video
`video_error_or_removed`	bool	Whether the video was removed or had an error
`description`	object	Description of the video (some missing values)
`category`	object	Category name (some missing values)
`country`	object	Country where the video was uploaded

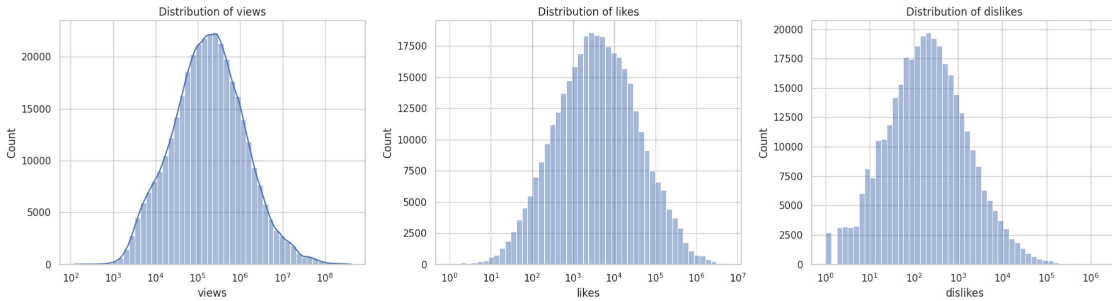
2.2 Handling NaN Data and Data Cleaning

- 2710 null values found, which deleted from dataset

2.3 Distributions

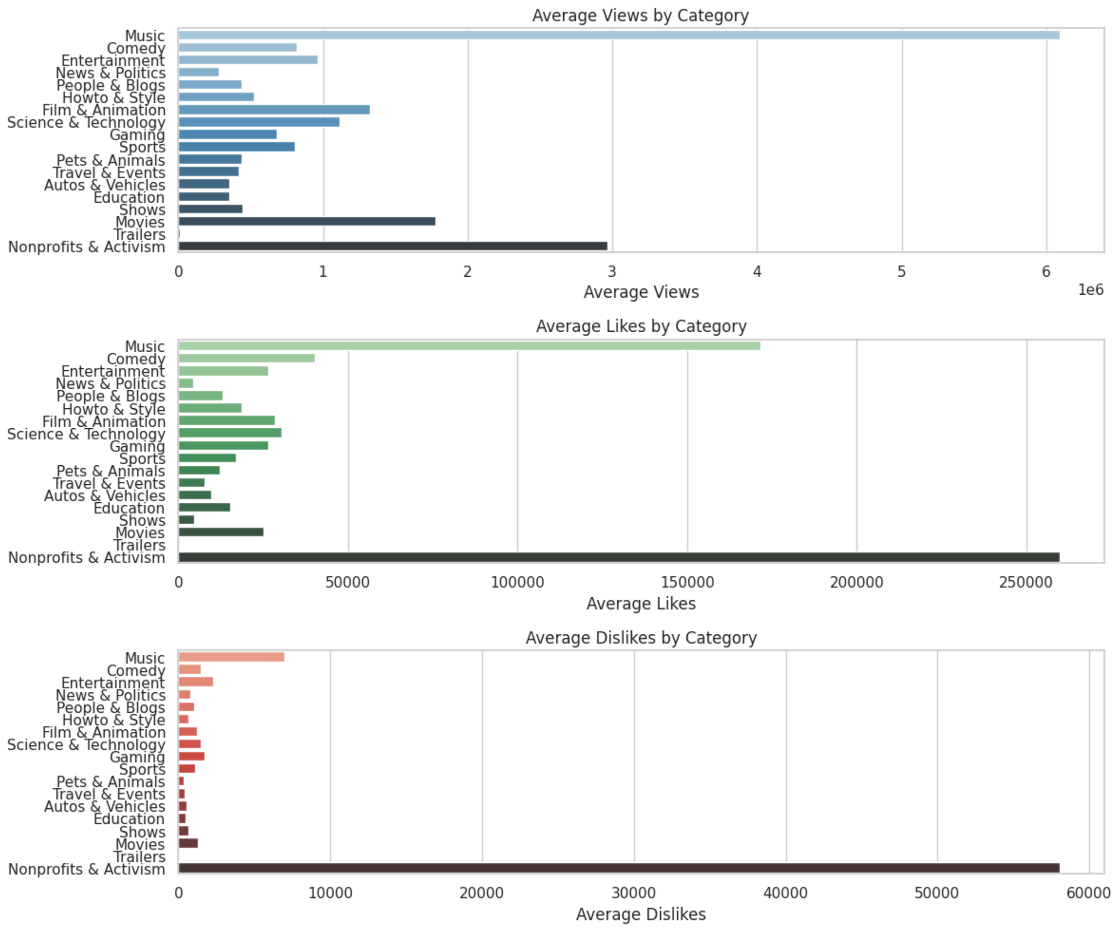
1. Log scale is useful for Skewed data, multiplicative growth (like YouTube views, income) and it Compresses large values and spreads small ones. Example: If we have views like [100, 200, 500, 1000000]: On linear scale: the last bar dominates. On log scale: [log10(100)=2, log10(200)=2.3, log10(500)=2.7, log10(1000000)=6] — more balanced view.

So log-scaling is not just scaling — it's a lens to reveal hidden structure in highly skewed data. For engagement metrics, it's almost always the best first look.[1]



It is resonable that dislike plot is skewed, because a video is less likely to receive a lot of dislikes.

2. Countries with closer cultural backgrounds tend to show similar statistics across categories. For example, in South Korea and Japan, the "Science & Technology" category receives more engagement and in Mexico, "Comedy". "Music" usually ranks first almost everywhere.



Globally, it is reasonable that the 'Nonprofits and Activism' category receives more feedback—both likes and dislikes—since it often reflects people's support or opposition to certain beliefs. The number of views remains highest in the Music category, which also makes sense.

2.4 Controversial Videos

We observe that metrics are generally higher for the universally liked cases, except for dislikes. The inference that can be drawn from this is that people may express their disagreement through dislikes when watching controversial videos. Of course, it's worth noting that this category was defined based on the dislike ratio.

Mann-Whitney U Test: statistic = 897053409.000, p-value = 0.00000

The Mann-Whitney U test with a p-value < 0.00001 confirms that the differences in distributions between the two groups are statistically significant, not due to random chance.

Inference: People might engage more deeply (watch, like, comment) with content they enjoy or agree with, while they use the dislike button more selectively for expressing disagreement — especially in controversial topics.

2.5 Tags

Inference: It seems that short title videos (0 to 20 characters) have more engagement ratio

2.6 Statistical Questions

2.6.1 Is there a significant association between the day of the week a video is published and its likelihood of trending?

We only have videos that already trended, so: We know which days are more common among trending videos. But We don't know how many total videos were uploaded on those days. So We can't compute true probabilities (likelihoods) !!!

2.6.2 Is there a significant difference in viewer engagement (likes-to-views ratio) across different video categories?

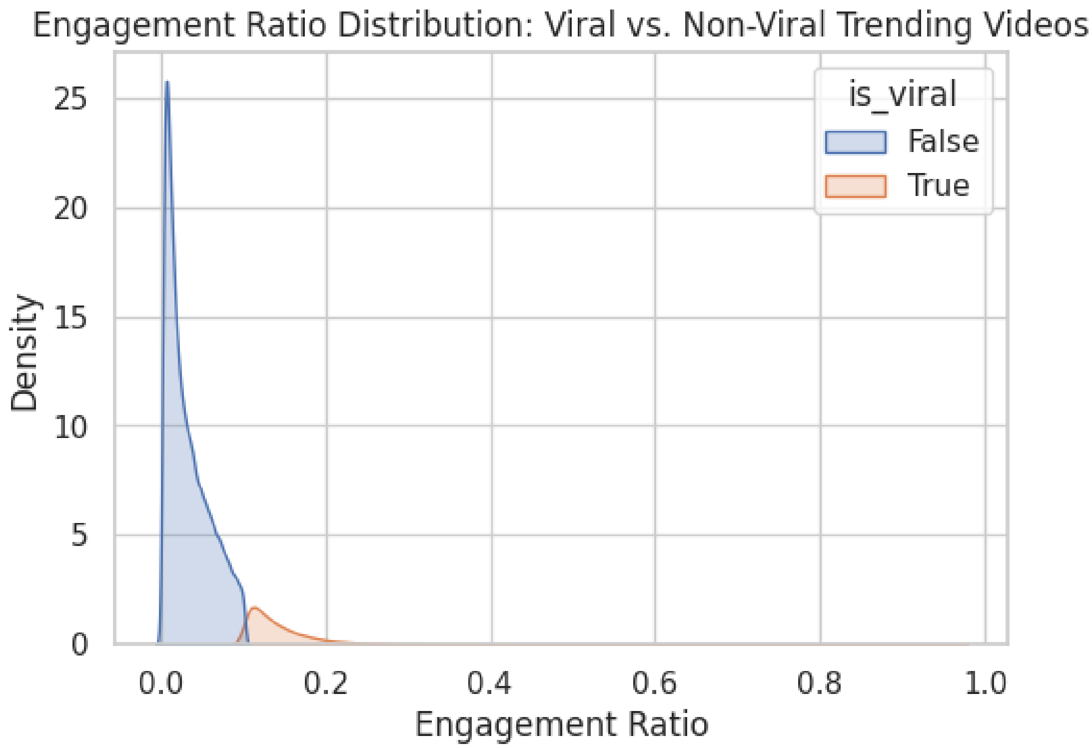
Kruskal-Wallis H-test: statistic = 44479.9468, p-value = 0.00000 The p-value is less than 0.05, reject the null hypothesis— implying a significant difference exists in engagement across categories. The boxplot helps to see which categories have higher/lower typical engagement.

2.7 virality: Questions Based On Story

All trending videos aren't created equal—some explode, others fizzle. What patterns can we find that explain why certain trending videos perform far better than others?

Since all videos are trending, we define virality using a measurable threshold: Top 10% in engagement ratio

Viral threshold for engagement_ratio: 0.10200935966734673 * False 318711 * True 35413



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2.8 Conclusion

The analysis of YouTube trending videos has revealed several interesting insights into the factors influencing video virality and engagement. Here are the key findings:

1. **Engagement Metrics and Distribution**:
   Engagement metrics such as views, likes, dislikes, and comments show a skewed distribution, with a small number of videos receiving disproportionately high engagement. The use of log-scale transformation helped reveal hidden patterns in these skewed data distributions, especially in engagement metrics like views.

2. **Video Categories and Viewer Engagement**:
   There is a significant difference in engagement across various video categories. The Kruskal-Wallis H-test revealed that certain categories, such as "Music," tend to receive significantly higher engagement compared to others like "Nonprofits and Activism." This highlights the cultural and thematic preferences that play a role in viewer interaction.

3. **Virality and Engagement Ratio**:

The concept of virality was defined using the top 10% engagement ratio, and a clear distinction was made between viral and non-viral videos. An analysis of title sentiment revealed a statistically significant difference in sentiment categories between viral and non-viral videos, with viral videos more likely to feature positive sentiments. This suggests that positive sentiment in titles may play a key role in driving engagement.

4. **The Effect of Tags on Virality**:
   The number of tags (num_tags) has a significant effect on video virality. The Mann-Whitney U Test confirmed that viral videos tend to use a higher number of tags, suggesting that including multiple relevant tags can increase the chances of a video becoming viral. However, more tags don't always guarantee virality, and careful selection of tags is still important.

5. **The Role of Publish Hour**:
   The analysis also explored whether the hour at which videos are published has an impact on their virality. It was found that viral videos tend to be posted at specific times of the day, with a more concentrated publishing window correlating with higher engagement.

6. **Clickbait Titles and Virality**:
   The study showed that clickbait-style titles, characterized by sensational phrases like "shocking" or "must watch," can have a significant impact on virality in some categories, particularly in "Entertainment," "News & Politics," and "People & Blogs." However, in many categories, clickbait titles were associated with lower viral success, suggesting that while these titles may draw initial attention, they do not always foster sustained engagement.

7. **Channel Frequency and Content Quality**:
   Channels that appear frequently in viral videos were identified, and it was observed that these channels do not necessarily post more often but tend to create better content that resonates more with viewers. This implies that content quality, rather than posting frequency alone, is a key factor in achieving viral success.

8. **Cross-Country Differences**:
   The analysis of cross-country data showed that cultural preferences influence what goes viral in different regions. For instance, "Music" videos consistently ranked highly globally, but in countries like South Korea and Japan, "Science & Technology" videos garnered more engagement, while "Comedy" was more popular in Mexico. This indicates that regional trends and cultural interests strongly shape video virality.

In summary, this analysis provides a deeper understanding of what makes YouTube videos go viral. Factors like engagement ratio, title sentiment, number of tags, the use of clickbait, and publishing time play crucial roles in determining whether a video will achieve viral success. Furthermore, regional preferences and content quality are significant drivers of video popularity across different countries. These insights can guide content creators in shaping their strategies to improve video engagement and increase the likelihood of virality.

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